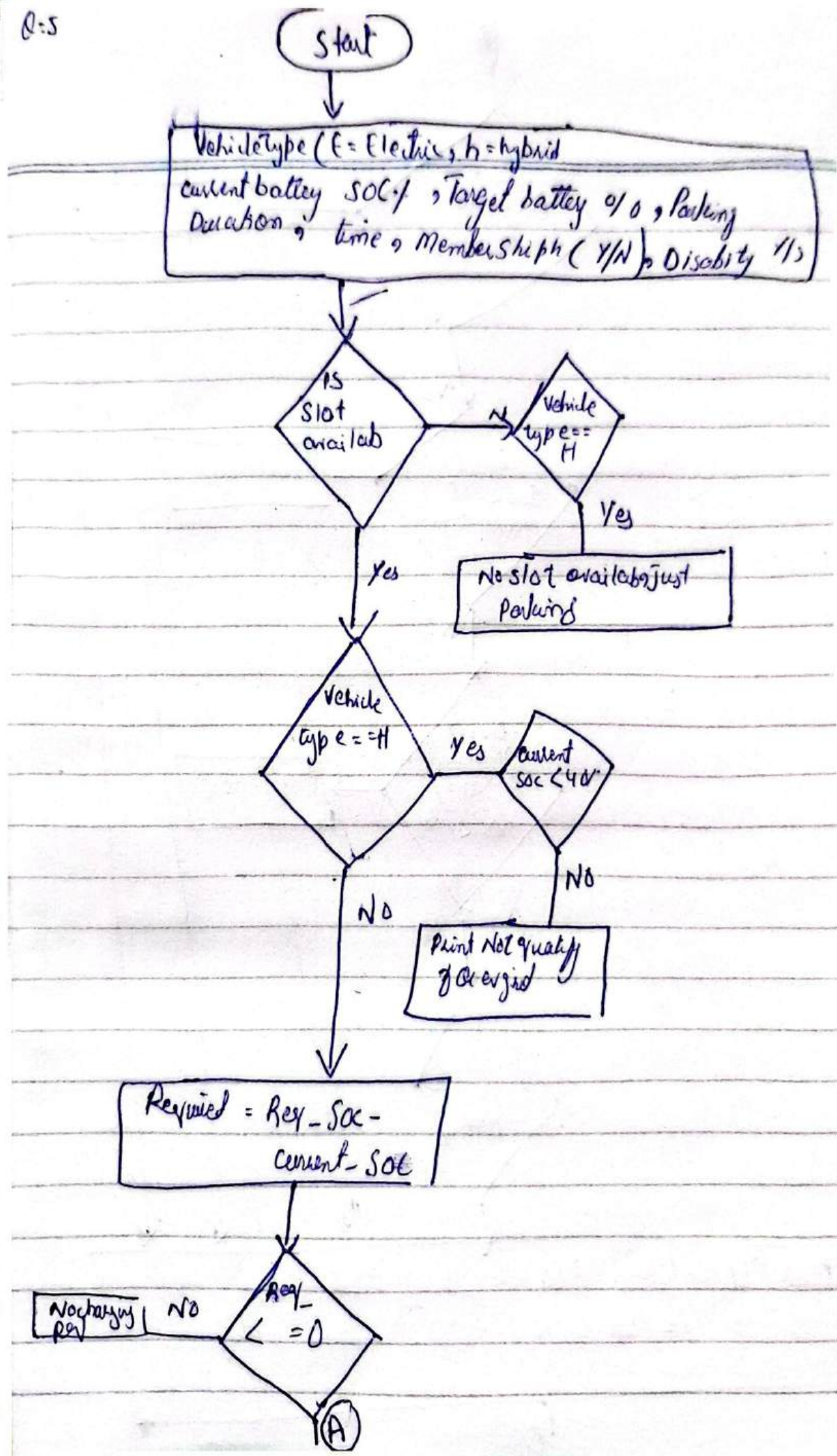
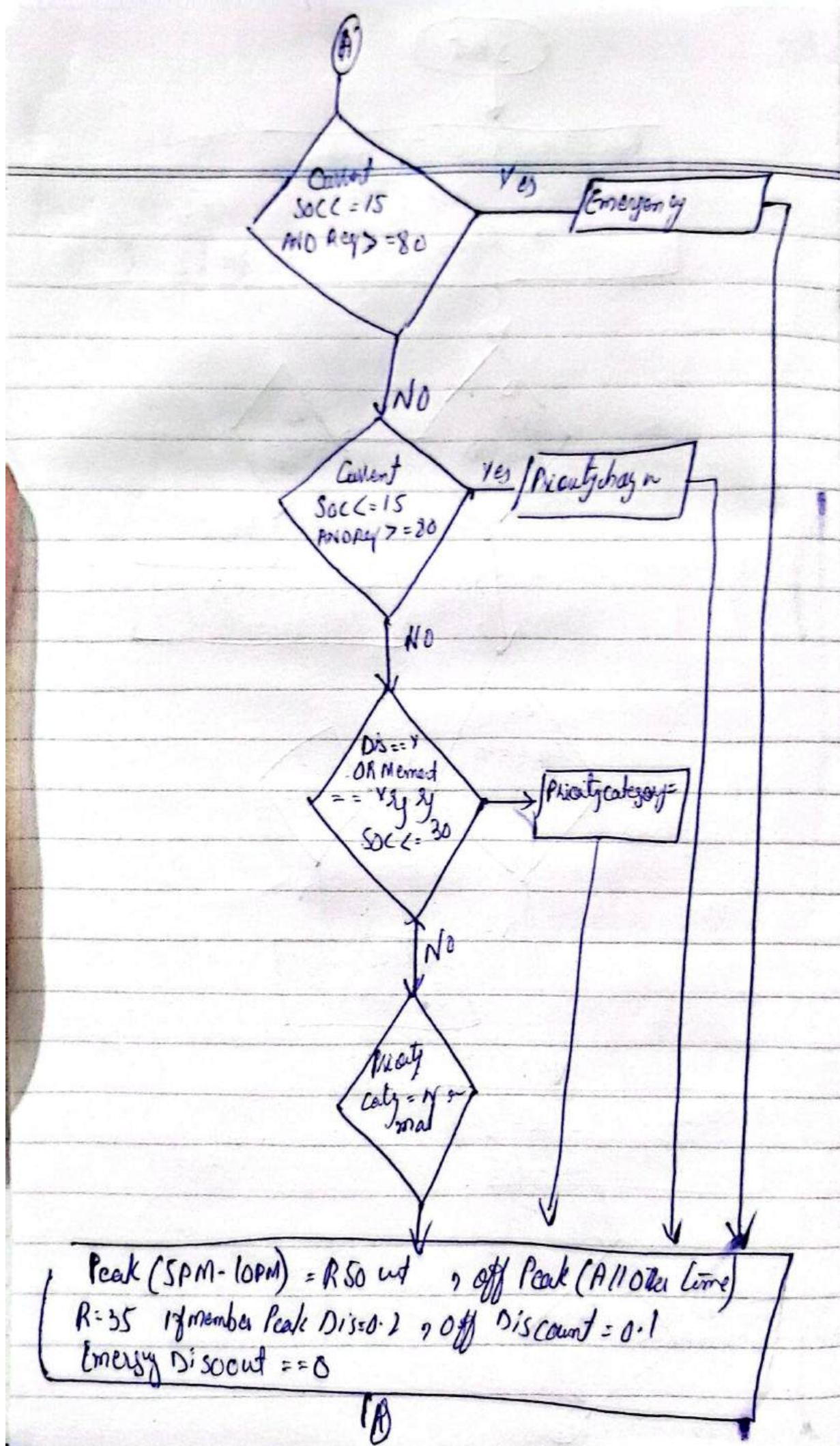


Q:5





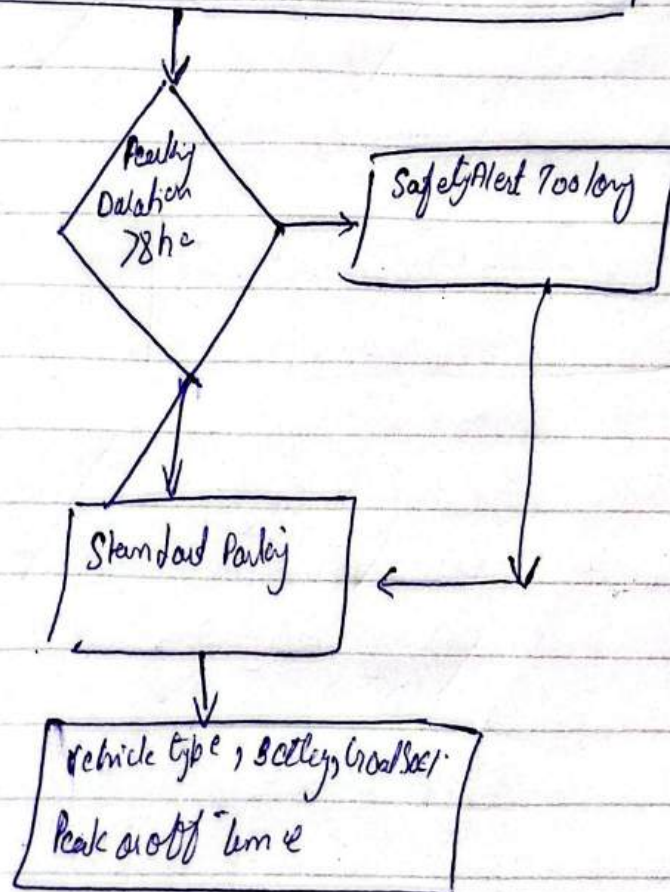
(A)

Parking time < 2 hour = Rs = 200,
2 to 5 hour = Rs = 400, > 5 hour Rs = 700

Else if member == Y apply 20% on

Base Parking fee

Final Balance = Net charging + Parking cost net



IPO Chart (Input - Processing - Output)

Input	Process	Output
• Type E/h • SOC, Avg SOC, (%) • Hours • Times • Member (Y/N) • Disabled (Y/N)	1. Check station availability by Vehicle eligibility. 2. Calculate req units to charge 3. Determine Priority Category 4. Compute energy fees Discount	- System/Errors msg • Assign Priority Status • Net charging cost • Net parking cost • Grand Total (Rs) • Stay of Duration
	• Parking charges with waives • Safety check	

Algorithm

- 1 Start by Read All input (Type, SOC, Req-SOC, Turning Member, Disabled, station-Avail)
- 2 If Station-Avail is 'N' Print error
(Parking for Hybrid only, "No Start fee (V)" & Jump to park)
- 3 If Type = H & SOC ≥ 40 Print Not Eligible
- 4 Calculate Req-units = Req-SOC - SOC. If Req ≤ 0 set it to zero No charge Required
- 5 If Emergency SOC = 15 & Req-SOC ≥ 800
- for Priority If Disabled == Y OR Member == Y & SOC ≤ 300
• Normal: All other qualify
- 6 Peak Window (5 PM - 10 PM) R = 50 unit Member dis = 10% = Emergency
Off Peak (discount = 20% • R = 35)
- 7 Parking cost (≤ 2 hr = 200, 2 to 5 hr R = 400, > 5 hr = 700 Rs)
Set parking fee = 0. Apply 20% if member == Y
- 8 If Hour > 8 , charging hour Set a long safe warn.
- 9 Data Invoice
- 10 End

Pseudocode

Start

Input Type, SOC, Rev-SOC, Hour, Turn, member,
Disabled, station-Avail

If station-Avail == 'N' then

 If Type == H then Print 'Parking only' Else 'No slot avail'

End If

2 Required^{units} = Rev-SOC - SOC

If Rev-unit <= 0 then

 Required = 0

 Print 'No charging required'

End If

If SOC <= 15 & Rev-SOC >= 80 Then

 Emergency

Else If Disabled == Y or member = Y & If

 SOC <= 30 then 'Priority charging'

Else

 Normal charging

End If

If Time $\geq 17:00$ AND Time $\leq 22:00$ then
Rate 50

If Member == Y And Is emergency == false Discount = 0.1
Net charging member == Y & Is emergency false Discount =
0.20 Else Discount = 0
End If

If Hours ≤ 2 then Base-Park = 200
Else If Hours ≤ 5 then Park = 400
Else Base Park = 700
End If

If Disabled == Y then Net-Parking = 0
Else member == then NetPark = Base-Park * 0.80
Else NetPark = Base Park
End If

Total - Payable = Net - charging + Net Parking
If Hour > 8 then warning = long stay alert
Else warning = "Standard duration"

End If

Output

Typ, SOC, Pay-Son, Priorities, en

End

PAC

Given Data

Type (E/H)
SOC, Reg - SOC (%)
Hours, Time,
Member n Disabled
Station Mail

Required Output

- System Access
- Assign Priority
- Category
- Net Chaying
- Grand Total

Processing

Check station availability
Hybrid eligibility
 $Unit = Reg - SOC - SOC$
 $if < 0$ set to 0

Total = Net chaying +
Net parking

Duration warning
If hour > 8

Condition

Hybrid must have
 $SOC < 40\%$ to

- Qualify
- No chaying
 - discount if Priority

• Parking fee is Rs. 0
if Disabled = 1